

NexStar Finance Enterprise Data Management Architecture and Implementation Plan

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DAT515- Enterprise Data Management

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January 12, 2026

Executive Summary

NexStar Finance, a rapidly expanding digital financial services company, is facing critical challenges resulting from fragmented data systems, inconsistent governance, and the absence of unified enterprise architecture. As the organization has grown across the United States, Latin America, and parts of Africa, its technology ecosystem has evolved in a decentralized manner, leading to siloed customer, transaction, and vendor information. This fragmentation has contributed to declining market share from 16.2% in 2021 to 12.8% in Q1 of the current year and has limited NexStar's ability to generate reliable enterprise analytics, meet regulatory expectations, and compete with more integrated digital banking providers.

The scope of this report is to assess NexStar's current enterprise architecture and data-management landscape, evaluate the organization's existing capabilities against industry standard frameworks, and identify operational impacts resulting from these gaps. The goal is to establish a foundation for a strategic data-architecture roadmap that supports scalability, global compliance, centralized analytics, and long-term competitiveness.

This document outlines the current state of NexStar's enterprise systems, evaluates alignment with principles from frameworks such as DAMA-DMBOK and TOGAF, and highlights deficiencies that hinder performance, governance, integration, and customer insights. These findings will inform us of the subsequent development of a comprehensive enterprise data strategy and an 18-month modernization plan.

Current State Assessment

Current Enterprise Architecture Overview

NexStar's enterprise architecture is a hybrid environment composed of multiple cloud platforms including AWS and Azure alongside legacy on premises systems inherited through acquisitions. These disparate systems operate independently, resulting in limited interoperability. Customer data resides in separate regional databases; transaction data is stored across isolated cloud instances; and cryptocurrency investment histories are maintained in stand-alone systems with minimal API-based integration.

This patchwork architecture lacks the foundational components of a cohesive enterprise architecture as described in the TOGAF Architecture Development Method, which emphasizes standardization, integration, and a unified technology vision across the enterprise (The Open Group, 2018). NexStar currently operates without a centralized data store, enterprise service bus, or customer master record.

Current Data Management Tools and Tech

NexStar's current data-management tools and technologies reveal significant gaps when evaluated against the DAMA-DMBOK framework. Several existing components are underperforming or unable to meet the company's operational demands. Data integration remains a core weakness, as current ETL tools cannot reliably manage high-volume, cross-regional data movement, leading to inconsistent refresh cycles and unreliable analytics. Data-quality management is similarly deficient; without a standardized quality program, the organization continues to struggle with duplicate customer profiles, inconsistent field structures, and unresolved transaction discrepancies. Metadata management presents another challenge, critical

metadata is either undocumented or manually maintained across teams, conflicting with DAMA expectations for proper governance, lineage tracking, and documentation standards (DAMA International, 2017). In addition to these shortcomings, NexStar is missing several foundational enterprise-data components entirely. The absence of a Master Data Management (MDM) solution prevents identity resolution and creates obstacles in fraud monitoring and delivering unified customer experience. Likewise, without a centralized data-governance office, the organization lacks formal oversight mechanisms necessary for ensuring compliance with global regulations such as GDPR, CCPA, and emerging regional banking requirements. NexStar also operates without an enterprise data warehouse, resulting in decentralized analytics that inhibit predictive modeling, customer segmentation, and enterprise reporting. Compounding these issues is the lack of a standardized API layer, which limits interoperability due to inconsistent development practices and missing version control. Collectively, these deficiencies show that NexStar's tools and processes fall short of industry-standard principles for integration, data quality, governance, and analytics readiness.

Operational Impact

These weaknesses in NexStar's data-management environment also have substantial operational impacts. The absence of centralized analytics prevents leadership from accessing real-time insights into customer behavior, market trends, or overall operational performance, reducing the company's ability to make informed strategic decisions and diminishing competitiveness in fast-moving fintech markets. Operational costs are elevated due to redundant systems, manual data reconciliation, and the increased labor required to compensate for weak integrations and inconsistent data quality. Compliance and security risks are also heightened, as the lack of cohesive governance exposes the company to potential audit failures, data-privacy violations,

and increased vulnerability to cyber threats, particularly concerning given NexStar's multi-regional footprint. Moreover, customers experience inconsistent service due to duplicate records, mismatched identity data, and irregular transaction-processing speeds across markets. Finally, NexStar's ability to scale is severely constrained; as the company grows, its current siloed and fragmented architecture cannot support rising transaction volumes or new service offerings without operational strain and performance degradation.

Future Impact

If left unaddressed, these limitations will hinder NexStar's ability to adopt advanced technologies such as AI-driven fraud detection, personalization engines, or blockchain-enhanced settlement systems. The company's market share is likely to continue declining as competitors leverage integrated data ecosystems to innovate more rapidly.

Past Impact

Historically, these data issues have already limited NexStar's ability to synthesize customer insights during expansion into African markets and have slowed the rollout of crypto-investment reporting features.

Current-State Assessment

NexStar Finance operates across multiple cloud platforms and inherited legacy systems, resulting in siloed customer, transaction, and vendor data. Although the company has invested heavily in mobile-first solutions, it lacks a centralized analytics environment, a unified customer master record, and standardized reporting capabilities. These limitations have contributed to

declining market share as competitors leverage more integrated enterprise data architectures.

Addressing these challenges requires both a well-structured relational data model and improvements in data governance and technology strategy.

Data Model

Relational and Logical Schema Overview

To address NexStar's core operational needs, customer management, transaction tracking, and vendor oversight, a three-table relational data model was developed. The model reflects a logical separation of entities while maintaining clear relationships that support reporting, analytics, and compliance.

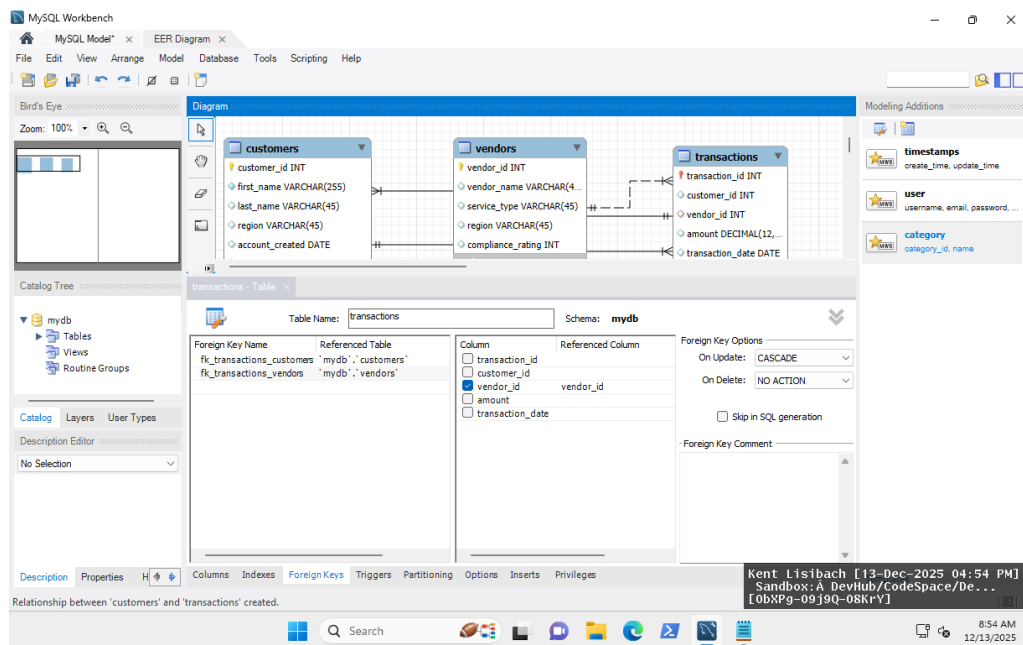


Figure 1 relational data model

The schema consists of the following tables:

- Customers
customer_id, first_name, last_name, region, account_created
- Transactions
transaction_id, customer_id (foreign key), vendor_id (foreign key), amount,
transaction_date
- Vendors
vendor_id, vendor_name, service_type, region, compliance_rating

Each table contains five attributes and ten sample records, ensuring the model meets the assignment's data completeness requirements. This data model supports NexStar's need to consolidate siloed systems by enabling centralized transaction processing, regional customer analytics, and vendor performance monitoring within a unified structure.

SQL Queries and Business Relevance

Ten SQL queries were developed using the proposed schema to demonstrate how NexStar can extract meaningful insights from its data. These queries incorporate aggregation functions, filtering, sorting, and joins across tables.

Examples of query use cases include:

- Counting customers by region to support market analysis
- Summing transaction amounts to evaluate financial volume

- Identifying minimum and maximum transaction values for risk assessment
- Filtering high-value transactions for compliance monitoring
- Joining customers, transactions, and vendors to create unified reporting views

Each query directly aligns with NexStar’s business needs, such as improving executive reporting, enabling fraud detection, supporting compliance audits, and enhancing customer segmentation. Screenshots of all ten queries and their results are included using the uCertify snipping tool, with brief explanations describing how each query supports the data model and organizational objectives

Customer_table

The screenshot displays the MySQL Workbench interface. The 'Query Editor' window shows the following SQL query:

```
1 USE mydb;
2 INSERT INTO customers (customer_id, first_name, last_name, region, account_created) VALUES
3 (1, 'A', 'A', 'US', '2021-01-01'),
4 (2, 'B', 'B', 'US', '2021-02-01'),
5 (3, 'C', 'C', 'LATAM', '2021-03-01'),
6 (4, 'D', 'D', 'LATAM', '2021-04-01'),
7 (5, 'E', 'E', 'AFRICA', '2021-05-01'),
8 (6, 'F', 'F', 'AFRICA', '2021-06-01'),
9 (7, 'G', 'G', 'US', '2021-07-01'),
10 (8, 'G', 'G', 'US', '2021-08-01'),
11 (9, 'I', 'I', 'AFRICA', '2021-09-01'),
12 (10, 'J', 'J', 'US', '2021-10-01');
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	08:57:16	USE mydb	0 row(s) affected	0.031 sec
2	08:58:38	SELECT region, COUNT(*) AS customer_count FROM customers GROUP BY r...	0 row(s) returned	0.032 sec / 0.000 sec
3	09:00:51	select * from customers LIMIT 0, 1000	0 row(s) returned	0.031 sec / 0.000 sec
4	09:35:03	Insert into customers (customer_id, first_name, last_name, region, account_created)	Error Code: 1064. You have an error in your SQL syntax; check the manual that...	0.140 sec
5	09:35:33	Insert into customers (customer_id, first_name, last_name, region, account_created)	Error Code: 1064. You have an error in your SQL syntax; check the manual that...	0.140 sec
6	09:37:20	Insert into customers (customer_id, first_name, last_name, region, account_created)	10 row(s) affected Records: 10 D...	0.140 sec

The error message in the output window is: "Error Code: 1064. You have an error in your SQL syntax; check the manual that...". The user's name and session information are visible at the bottom right: "Kent [13-Dec-2025 05:37 PM] Sandbox-A: DevHub/CodeSpace/De... [0bXPg-09j9Q-08KrY]".

Vendors table

MySQL Workbench

MySQL Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Query 1

```

1 USE mydb;
2 INSERT INTO vendors (vendor_id, vendor_name, service_type, region, compliance_rating) VALUES
3 (1, 'V1', 'PAY', 'US', 5),
4 (2, 'V2', 'PAY', 'US', 4),
5 (3, 'V3', 'Crypto', 'LATAM', 4),
6 (4, 'V4', 'Crypto', 'LATAM', 3),
7 (5, 'V5', 'Crypto', 'LATAM', 3),
8 (6, 'V6', 'Remite', 'AFRICA', 4),
9 (7, 'V7', 'Pay', 'US', 5),
10 (8, 'V8', 'Crypto', 'LATAM', 3),
11 (9, 'V9', 'Remite', 'AFRICA', 2),
12 (10, 'V10', 'Pay', 'US', 4);
13

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
2	08:58:38	SELECT region, COUNT(*) AS customer_count FROM customers GROUP BY region	0 row(s) returned	0.032 sec / 0.000 sec
3	09:00:51	select * from customers LIMIT 0, 1000	0 row(s) returned	0.031 sec / 0.000 sec
4	09:35:03	insert into customers (customer_id, first_name, last_name, region, account_creation_date) values (1, 'John', 'Doe', 'US', '2024-01-01')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.140 sec
5	09:35:33	insert into customers (customer_id, first_name, last_name, region, account_creation_date) values (2, 'Jane', 'Doe', 'US', '2024-01-01')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
6	09:37:20	insert into customers (customer_id, first_name, last_name, region, account_creation_date) values (3, 'John', 'Doe', 'US', '2024-01-01')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.187 sec
7	09:45:08	insert into vendors (vendor_id, vendor_name, service_type, region, compliance_rating) values (1, 'V1', 'PAY', 'US', 5)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.141 sec
8	09:50:48	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (1, 1, 1, 100, '2024-01-01')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
9	09:54:09	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (2, 2, 2, 200, '2024-01-02')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
10	09:55:59	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (3, 3, 3, 300, '2024-01-03')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
11	09:59:27	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (4, 4, 4, 400, '2024-01-04')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

Object Info Session

9:45 AM 12/13/2025

Transactions

MySQL Workbench

MySQL Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Query 1

```

1 USE mydb;
2 INSERT INTO transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) VALUES
3 (1, 1, 1, 100, '2024-01-01'),
4 (2, 2, 2, 200, '2024-01-02'),
5 (3, 3, 3, 300, '2024-01-03'),
6 (4, 4, 4, 400, '2024-01-04'),
7 (5, 5, 5, 500, '2024-01-05'),
8 (6, 6, 6, 600, '2024-01-06'),
9 (7, 7, 7, 700, '2024-01-07'),
10 (8, 8, 8, 800, '2024-01-08'),
11 (9, 9, 9, 900, '2024-01-09'),
12 (10, 10, 10, 1000, '2024-01-10');
13

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output

#	Time	Action	Message	Duration / Fetch
6	09:37:20	insert into customers (customer_id, first_name, last_name, region, account_creation_date) values (1, 'John', 'Doe', 'US', '2024-01-01')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.187 sec
7	09:45:08	insert into vendors (vendor_id, vendor_name, service_type, region, compliance_rating) values (1, 'V1', 'PAY', 'US', 5)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.141 sec
8	09:50:48	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (1, 1, 1, 100, '2024-01-01')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
9	09:54:09	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (2, 2, 2, 200, '2024-01-02')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
10	09:55:59	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (3, 3, 3, 300, '2024-01-03')	Error Code: 1064. You have an error in your SQL syntax; check the manual that	0.000 sec
11	09:59:27	insert into transactions (transaction_id, customer_id, vendor_id, amount, transaction_date) values (4, 4, 4, 400, '2024-01-04')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

Object Info Session

9:58 AM 12/13/2025

Query 1: (COUNT)

MySQL Workbench

MySQ Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Status and System Variables
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1

```

1 USE mydb;
2 select region, count(*) AS customer_count
3 FROM customers
4 GROUP BY region;
5

```

Result Grid

region	customer_count
US	5
LATAM	2
AFRICA	3

Result 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	09:50:48	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	Error Code: 1064. You have an error in your SQL syntax; check the manual th...	0.000 sec
9	09:54:09	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	Error Code: 1064. You have an error in your SQL syntax; check the manual th...	0.000 sec
10	09:55:59	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	Error Code: 1064. You have an error in your SQL syntax; check the manual th...	0.000 sec
11	09:58:27	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.109 sec
12	10:01:54	USE mydb	0 row(s) affected	0.000 sec
13	10:01:54	select region, count(*) AS customer_count FROM customers GROUP BY reg...	3 row(s) returned	0.047 sec / 0.015 sec

10:01 AM
12/13/2025

Query 2: (COUNT, GROUP BY)

MySQL Workbench

MySQ Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1

```

1 USE mydb;
2 SELECT SUM(amount) AS total_amount
3 FROM transactions;
4

```

Result Grid

total_amount
5500.00

Result 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	09:55:59	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	Error Code: 1064. You have an error in your SQL syntax; check the manual th...	0.000 sec
11	09:58:27	insert into transactions (transaction_id, customer_id, vendor_id, amount, trans...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.109 sec
12	10:01:54	USE mydb	0 row(s) affected	0.000 sec
13	10:01:54	select region, count(*) AS customer_count FROM customers GROUP BY reg...	3 row(s) returned	0.047 sec / 0.015 sec
14	10:03:25	USE mydb	0 row(s) affected	0.000 sec
15	10:03:25	SELECT SUM(amount) AS total_amount FROM transactions LIMIT 0. 1000	1 row(s) returned	0.000 sec

10:03 AM
12/13/2025

Query 3: Minimum transaction amount (MIN)

MySQL Workbench

MySQL Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1 x

```
1 USE mydb;
2 SELECT MIN(amount) AS smallest_transaction
3 FROM transactions;
```

Result Grid

smallest_transaction
100.00

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	10:01:54	USE mydb	0 row(s) affected	0.000 sec
13	10:01:54	select region, count(*) AS customer_count FROM customers GROUP BY regi...	3 row(s) returned	0.047 sec / 0.015 sec
14	10:03:25	USE mydb	0 row(s) affected	0.000 sec
15	10:03:25	SELECT SUM(amount) AS total_amount FROM transactions LIMIT 0, 1000	1 row(s) returned	0.046 sec / 0.000 sec
16	10:05:47	USE mydb	0 row(s) affected	0.000 sec
17	10:05:47	SELECT MIN(amount) AS smallest_transaction FROM transactions LIMIT 0, 1...	1 row(s) returned	0.062 sec / 0.000 sec

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10:05 AM
12/13/2025

Query 4: Maximum transaction amount (MAX)

MySQL Workbench

MySQL Model (milestone 2.mwb) x EER Diagram x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1 x

```
1 USE mydb;
2 SELECT MAX(amount) AS largest_transaction
3 FROM transactions;
```

Result Grid

largest_transaction
1000.00

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	10:03:25	USE mydb	0 row(s) affected	0.000 sec
15	10:03:25	SELECT SUM(amount) AS total_amount FROM transactions LIMIT 0, 1000	1 row(s) returned	0.046 sec / 0.000 sec
16	10:05:47	USE mydb	0 row(s) affected	0.000 sec
17	10:05:47	SELECT MIN(amount) AS smallest_transaction FROM transactions LIMIT 0, 1...	1 row(s) returned	0.062 sec / 0.000 sec
18	10:07:03	USE mydb	0 row(s) affected	0.000 sec
19	10:07:03	SELECT MAX(amount) AS largest_transaction FROM transactions LIMIT 0, 1...	1 row(s) returned	0.062 sec / 0.000 sec

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Sandbox: A DevHub/CodeSpace/De...
[0bXPg-09j9Q-08KrY]

10:07 AM
12/13/2025

Query 5: Transactions over 500 (WHERE filter)

MySQL Workbench interface showing Query 1 with the following SQL:

```
1 • USE mydb;
2 • SELECT transaction_id, amount
3 • FROM transactions
4 • WHERE amount > 500;
```

The Result Grid displays the following data:

transaction_id	amount
6	600.0
7	700.0
8	800.0
9	900.0
10	1000.0
11	1100.0
12	1200.0

The Output pane shows the Action Output table:

#	Time	Action	Message	Duration / Fetch
16	10:05:47	USE mydb	0 row(s) affected	0.000 sec
17	10:05:47	SELECT MIN(a...	1 row(s) returned	0.062 sec / 0.000 sec
18	10:07:03	USE mydb	0 row(s) affected	0.000 sec
19	10:07:03	SELECT MAX(a...	1 row(s) returned	0.000 sec / 0.000 sec
20	10:08:23	SELECT transaction_id, amount FROM transactions WHERE amount > 500	7 row(s) returned	0.062 sec / 0.000 sec
21	10:08:23	USE mydb	0 row(s) affected	0.000 sec

Query 6: Transactions ordered by amount (ORDER BY)

MySQL Workbench interface showing Query 1 with the following SQL:

```
1 • USE mydb;
2 • SELECT transaction_id, amount
3 • FROM transactions
4 • ORDER BY amount DESC;
```

The Result Grid displays the following data (ordered by amount):

transaction_id	amount
10	1000.00
9	900.00
8	800.00
7	700.00
6	600.00
5	500.00
4	400.00
3	300.00
2	200.00
1	100.00

The Output pane shows the Action Output table:

#	Time	Action	Message	Duration / Fetch
18	10:07:03	USE mydb	0 row(s) affected	0.000 sec
19	10:07:03	SELECT MAX(amount) AS largest_transaction FROM transactions LIMIT 0, 1	1 row(s) returned	0.000 sec / 0.000 sec
20	10:08:23	USE mydb	0 row(s) affected	0.000 sec
21	10:08:23	SELECT transaction_id, amount FROM transactions WHERE amount > 500	7 row(s) returned	0.062 sec / 0.000 sec
22	10:09:28	USE mydb	0 row(s) affected	0.000 sec
23	10:09:28	SELECT transaction_id, amount FROM transactions ORDER BY amount DESC	10 row(s) returned	0.062 sec / 0.000 sec

Query 7: Join customers and transactions

MySQL Workbench

Query 1

```

1 • USE mydb;
2 • SELECT c.first_name, t.amount
3 • FROM customers c
4 • JOIN transactions t ON c.customer_id = t.customer_id;

```

Result Grid

first_name	amount
A	100.00
B	200.00
C	300.00
D	400.00
E	500.00
F	600.00
G	700.00

Output

#	Time	Action	Message	Duration / Fetch
24	10:10:45	USE mydb	0 row(s) affected	0.000 sec
25	10:10:45	SELECT c.first_name, t.amount FROM customers c JOIN transactions t ON c...	Error Code: 1054. Unknown column 't.customer' in 'on clause'	0.047 sec
26	10:11:18	USE mydb	0 row(s) affected	0.000 sec
27	10:11:18	SELECT c.first_name, t.amount FROM customers c JOIN transactions t ON c...	Error Code: 1054. Unknown column 't.customer' in 'on clause'	0.000 sec
28	10:11:40	USE mydb	0 row(s) affected	0.000 sec
29	10:11:40	SELECT c.first_name, t.amount FROM customers c JOIN transactions t ON c...	10 row(s) returned	

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Sandbox: A DevHub/CodeSpace/De...
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Query 8: JOIN vendors, total amount per customer

MySQL Workbench

Query 1

```

1 • USE mydb;
2 • SELECT c.customer_id, SUM(t.amount) AS total_spent
3 • FROM customers c
4 • JOIN transactions t ON c.customer_id = t.customer_id
5 • GROUP BY c.customer_id;

```

Result Grid

customer_id	total_spent
1	100.00
2	200.00
3	300.00
4	400.00
5	500.00
6	600.00
7	700.00

Output

#	Time	Action	Message	Duration / Fetch
31	10:13:17	SELECT c.customer_id, SUM(t.amount) AS total_spent FROM customers c J...	Error Code: 1630. FUNCTION mydb.SUM does not exist. Check the Function...	0.093 sec
32	10:14:13	SELECT c.customer_id, SUM(t.amount) AS total_spent FROM customers c J...	Error Code: 1064. You have an error in your SQL syntax; check the manual th...	0.000 sec
33	10:14:39	USE mydb	0 row(s) affected	0.000 sec
34	10:14:39	SELECT c.customer_id, SUM(t.amount) AS total_spent FROM customers c J...	Error Code: 1630. FUNCTION mydb.SUM does not exist. Check the Function...	0.000 sec
35	10:14:56	USE mydb	0 row(s) affected	0.000 sec
36	10:14:56	SELECT c.customer_id, SUM(t.amount) AS total_spent FROM customers c J...	10 row(s) returned	

Kent Lisibach [13-Dec-2025 06:15 PM]
Sandbox: A DevHub/CodeSpace/De...
[0bXPg-09J9Q-08KrY]

Query 9: Count transactions per vendor (JOIN + COUNT)

MySQL Workbench interface showing Query 9: Count transactions per vendor (JOIN + COUNT). The query is:

```

1. USE mydb;
2. SELECT v.vendor_name, COUNT(*) AS txn_count
3. FROM vendors v
4. JOIN transactions t ON v.vendor_id = t.vendor_id
5. GROUP BY v.vendor_name;

```

The result grid shows 7 rows of data:

vendor_name	txn_count
V1	1
V2	1
V3	1
V4	1
V5	1
V6	1
V7	1

The output pane shows an error: "Error Code: 1054. Unknown column 'v.vendor_name' in field list".

Query 10: High-value transactions with names (JOIN + WHERE)

MySQL Workbench interface showing Query 10: High-value transactions with names (JOIN + WHERE). The query is:

```

1. USE mydb;
2. SELECT c.first_name, v.vendor_name, t.amount
3. FROM transactions t
4. JOIN customers c ON t.customer_id = c.customer_id
5. JOIN vendors v ON t.vendor_id = v.vendor_id
6. WHERE t.amount > 500;

```

The result grid shows 6 rows of data:

first_name	vendor_name	amount
F	V6	600.00
G	V7	700.00
G	V	800.00
I	V9	900.00
J	V10	1000.00

The output pane shows an error: "Error Code: 1054. Unknown column 'v.vendor_name' in field list".

Technology Recommendations

Cloud-Native Data Warehouse

A cloud-native data warehouse (such as Snowflake or BigQuery) is recommended to centralize NexStar's structured and semi-structured data. Cloud-native data warehouses support elastic scalability, separation of storage and computing, and real-time analytics, making them well suited for organizations with rapidly growing and geographically distributed data environments (Kimball & Ross, 2013; Gartner, 2023).

Benefits:

This technology enables scalable storage, near real-time analytics, and centralized reporting across regions. It supports mobile and API-generated data and improves decision-making through faster, more consistent analytics.

Challenges:

Implementation requires data migration, updated governance standards, and staff training to ensure effective adoption.

Master Data Management (MDM) Platform

An MDM solution is recommended to establish a single, trusted customer master record across NexStar's global operations.

Benefits:

MDM reduces data duplication, improves data quality, enhances fraud detection, and enables more accurate customer insights.

Challenges:

Successful implementation requires strong governance, clearly defined data ownership, and compliance with regional privacy regulations.

Management Recommendations

NexStar should adopt data management best practices aligned with frameworks such as DAMA-DMBOK, emphasizing data governance, metadata management, and information lifecycle management. Standardizing data definitions, ownership, and quality rules will ensure consistency across structured and unstructured data sources, including transaction logs, support communications, and mobile application data. Formal data governance frameworks emphasize accountability, metadata management, and information lifecycle controls as critical enablers of scalable and ethical data usage across the enterprise (DAMA International, 2017).

Impact on Operations and Stakeholders

These recommendations will improve operational efficiency by reducing manual reconciliation, enabling unified analytics, and strengthening regulatory compliance. Internal stakeholders will benefit from improved reporting and decision-making capabilities, while external customers will experience more secure, reliable, and personalized financial services. Implementing new technologies will require updates to management strategies, particularly in governance and access control, to maximize value and ensure sustainability.

Ethical and Regulatory Considerations

NexStar must comply with global privacy and financial regulations, including GDPR and CCPA. Ethical considerations include protecting sensitive financial data, restricting access through role-based controls, encrypting customer information, and maintaining transparency in data usage. Strong governance and audit controls are essential to maintain trust and safeguard proprietary information.

Implementation Plan

If NexStar Finance accepts all recommended technology and management improvements, implementation should follow a phased, 18-month rollout to balance speed, risk, and organizational readiness. This approach allows NexStar to deliver incremental value while minimizing disruption to ongoing operations.

Phase 1: Governance and Strategic Foundation (Months 1–3)

The initial phase focuses on establishing enterprise governance and management structures. NexStar should create a formal Data Governance Office (DGO) aligned with DAMA-DMBOK principles, assigning executive sponsors, data owners, and data stewards across business domains (DAMA International, 2017). Key activities include defining enterprise data standards, metadata requirements, data-quality rules, and security policies.

This phase requires significant management attention, as executive leadership must enforce cross-departmental participation and ensure accountability. A major tradeoff is the upfront time

investment from leadership; however, this governance foundation is essential to prevent future data silos and compliance risks.

Phase 2: Core Technology Deployment (Months 4–9)

In Phase 2, NexStar implements its cloud-native enterprise data warehouse to centralize structured and semi-structured data from existing cloud platforms and legacy systems. Data integration pipelines should be established to ingest customer, transactions, and vendor data into the unified relational model recommended in earlier milestones.

This phase requires technical expertise in cloud engineering, data integration, and security. Tradeoffs include short-term increases in infrastructure costs and staff training requirements. However, consolidation significantly reduces long-term operational inefficiencies caused by duplicate systems and manual reconciliation. Centralized reporting and analytics capabilities also begin delivering immediate value to leadership and compliance teams.

Phase 3: Master Data Management and Analytics Expansion (Months 10–15)

The third phase focuses on deploying a Master Data Management (MDM) solution to create a single, trusted customer master record across regions. MDM resolves duplicate identities, improves fraud detection, and enables consistent regulatory reporting. Analytics capabilities can be expanded during this phase to support customer segmentation, transaction monitoring, and vendor performance analysis.

This phase requires ongoing collaboration between business and technical teams. The primary tradeoff is increased governance overhead, as data stewardship becomes a continuous

responsibility. However, the benefit is improved data accuracy, customer trust, and competitive positioning.

Phase 4: Optimization and Sustainability (Months 16–18)

The final phase emphasizes optimization, automation, and long-term sustainability. NexStar should implement continuous data-quality monitoring, metadata lineage tracking, and lifecycle management policies. Organization-wide training programs should be introduced to improve data literacy and ensure adoption of the new architecture.

Financial Impact and Cost-Effectiveness

The proposed architecture requires upfront investment but delivers strong long-term cost efficiency. Estimated costs include cloud infrastructure and storage, MDM licensing, integration tools, and training. While initial expenses may increase operational budgets in the short term, these costs are offset by reduced system redundancy, lower compliance risk, improved fraud detection, and faster decision-making.

Centralized analytics and governance also support revenue growth by enabling more accurate customer insights, improved personalization, and faster market responsiveness. Overall, the architecture represents a cost-effective investment aligned with NexStar's long-term growth strategy.

Ethical Standards and Data Protections

Because NexStar operates in the financial services sector, ethical data management is critical. The organization must ensure compliance with global privacy regulations while upholding principles of fairness, transparency, and accountability. Customer financial and cryptocurrency data must be protected through strong encryption, role-based access controls, and audit logging.

Ethical considerations also include transparency in analytics and automated decision-making processes, particularly those related to fraud detection or risk assessment. NexStar must ensure that data usage does not introduce bias or unfair treatment across regions and demographics. Strong governance structures help enforce ethical standards and maintain customer trust.

Conclusion

NexStar Finance's rapid global expansion has outpaced the maturity of its enterprise data architecture, leaving the organization constrained by fragmented systems, inconsistent governance, and limited analytical capability. This report demonstrated that NexStar's current environment fails to meet the expectations of industry-standard frameworks such as DAMA-DMBOK and TOGAF, resulting in operational inefficiencies, elevated compliance risk, and reduced competitive agility. Through a structured assessment of NexStar's architecture, data tools, and governance gaps, the company cannot continue to scale effectively without a fundamental modernization of its data management strategy.

The proposed enterprise data model, cloud-native data warehouse, and Master Data Management platform establish the technical foundation needed to unify customer, transaction, and vendor data into a single, trusted enterprise view. When combined with formal data governance, metadata management, and data-quality controls, NexStar gains the ability to

support real-time analytics, regulatory reporting, fraud detection, and strategic decision-making across all regions. These improvements directly address NexStar's historical challenges in market expansion, customer identity resolution, and reporting accuracy, while also positioning the organization for future innovation.

Looking forward, the implementation of this architecture enables NexStar to adopt advanced analytical tools such as machine learning–based fraud detection, predictive customer behavior modeling, and automated compliance monitoring. As data becomes more reliable, integrated, and governed, NexStar can leverage additional technologies such as AI-driven risk scoring, real-time transaction monitoring, and advanced customer personalization platforms. These capabilities will not only improve financial performance but also enhance customer trust and regulatory standing in an increasingly data-driven fintech environment.

Ultimately, by aligning its enterprise data strategy with modern frameworks and technologies, NexStar Finance transforms data from a fragmented operational liability into a strategic asset. The proposed roadmap allows the organization to scale securely, compete more effectively, and sustain long-term growth in global digital financial services.

References

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The Open Group. (2018). *TOGAF® standard, version 9.2*. The Open Group.